No.	Status	Date
QSCP-IP-ATU100 Rev 3	Issued	2018-12-14

Piping Inspection Plan - ATU-100 (RBI)

Scope and Objectives

This InspectionPlan defines the integrity management strategy for the piping systems within the ATU-100 at the Hamilton Plant. The document implements QSCP-CIS-001 Rev 0, transitioning the unit from a time-based inspection regime to a risk-based methodology. This revision supersedes QSCP-IP-ATU100 Rev 2 and incorporates the findings of the recent risk assessment to optimize the monitoring of active degradation mechanisms.

The primary objective is to ensure the continued mechanical integrity of 18 process lines through a rationalized network of 150 Condition Monitoring Locations (CMLs). Jennifer Walsh, as the Inspection Coordinator, is responsible for the execution of this plan, ensuring that all thickness data and Fitness-For-Service assessments comply with the requirements for qualified engineering review.

Risk-Based Inspection Intervals

Inspection frequencies have been established based on QSCP-RBI-ATU100-2018. A 3-year interval is mandated for high-risk services, specifically Class 1 sour gas, the regenerator overhead, and rich amine systems. All other process lines within the unit are assigned a 6-year inspection interval. These frequencies are designed to remain within the maximum code limits while addressing the specific corrosivity of the amine treating process.

The reflux lines are managed as a single corrosion loop. Integrity for this loop is assessed based on the loop-average wall margin and corrosion rate rather than individual component analysis. This approach allows for a statistically representative assessment of the piping circuit's condition while maintaining compliance with the corporate integrity standard.

Inspection Methodology and CML Management

The CML count has been rationalized to 150 points across the 18 lines to focus on areas of highest vulnerability. A full re-baseline survey of all lines is scheduled for October 2019 to establish a new data foundation under the revised standard. For A312 TP316 stainless steel piping, ultrasonic thickness measurements must be performed using the specific stainless-steel velocity setting to ensure accuracy.

Corrosion rates are determined by selecting the larger of the long-term rate or the short-term rate, provided the latest readings are at least 3 years apart. Any reading falling below the retirement thickness (t-min) requires the immediate issuance of an NCR. Remaining life calculations must demonstrate that the next inspection is scheduled at no more than half the remaining life.

Deferral Justification and Schedule

A one-time deferral has been granted for the Class 1 sour gas, regenerator overhead, and rich amine surveys originally due in February 2019. These inspections are now aligned with the October 2019 re-baseline survey. This deferral is technically justified by the review of 2016 data, which confirms that the remaining life at every CML is more than twice the duration of the deferral period.

Jennifer Walsh shall oversee the transition to the October 2019 schedule, ensuring that no integrity limits are exceeded during the extension. The CUI program remains active for all insulated lines operating within the CUI temperature range, requiring profile RT or insulation windows every 5 years as established in the previous integrity framework.

Sign-off	Name	Function	Date	Signature
Prepared	Jennifer Walsh	Inspection Coordinator, API 570 inspector	2018-12-14	_____